DiskGuard: Explainable, Agent-Assisted Hard-Drive Failure Management Evaluated at True Prevalence

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***Abstract*—Hard-drive failures are rare on any one day but frequent across a large fleet, and published prediction results are often reported on rebalanced test data that hide how many alerts an operator would actually receive. We present DiskGuard, an end-to-end failure-management system built on more than twelve years of Backblaze SMART telemetry (681.7 million drive-days). A five-seed LightGBM ensemble over 212 SMART-derived features predicts failure within 14 days. Evaluated at the true positive rate of 0.047 percent on a 15-month held-out period, it reaches a drive-day AUPRC of 0.084 (179 times the base rate) and a drive-level AUPRC of 0.600. Re-scoring the same predictions on a balanced test set would report an AUPRC of 0.917, an elevenfold inflation. Recall-calibrated risk tiers place 66.4 percent of failing drives in a HIGH tier with 57 percent precision. A survival model supplies an urgency ranking instead of an unreliable day count, TreeSHAP attributions are mapped to root causes and actions, and a tool-using language-model agent drafts maintenance tickets that a human must approve, with safety rules enforced in code and verified on 34,794 drives and ten adversarial scenarios. On six months of 2026 data published after the model was frozen, recall is maintained but AUPRC falls to 0.061; the decline is gradual and driven by growing false alarms, and a pre-registered retraining on more recent data recovers AUPRC to 0.078. A fixed daily alert budget then beats a fixed threshold, raising the precision of new alerts by 14 percent with 22 percent fewer alerts, and a drift monitor replayed on the frozen model would have called for retraining six months before the decline was measured.**

Keywords—hard disk drive failure prediction, SMART, gradient boosting, class imbalance, explainability, concept drift, language-model agents

I. Introduction

Hard disk drives remain the dominant medium for bulk storage in large datacenters, and they fail. Field studies report annual replacement rates of a few percent, well above datasheet figures [1], so at fleet scale failures arrive daily, and each one costs a rebuild and a window of reduced redundancy. Predictive maintenance, which scores every drive every day and replaces those likely to fail, is the natural response, and SMART (Self-Monitoring, Analysis and Reporting Technology) telemetry is its standard input [2], [3], [4].

Failure within a short horizon is a rare event. In our test period only 0.047 percent of drive-days are positive, so a model that always predicts healthy is 99.95 percent accurate. Accuracy, and to a lesser degree ROC-AUC, say little about whether the top of the ranking, where an operator actually looks, is any good [5]. Results computed on class-balanced test sets change the base rate and cannot be compared with deployment. A probability alone also does not tell an operator which drive to replace first, why it is at risk, or what to do.

We build and evaluate a complete system that answers four operational questions for every drive in a 352,000-drive fleet: will it fail soon, how urgent is it, why, and what should be done. Our contributions are:

- Failure prediction evaluated on 144.7 million held-out drive-days at true prevalence, with drive-clustered confidence intervals, and a measurement of how much common evaluation shortcuts inflate the same model's metrics (up to elevenfold in AUPRC).
- Recall-calibrated risk tiers fixed on validation, an urgency ranking from a survival model in place of an unreliable day count, and TreeSHAP explanations mapped to tier-aware actions.
- An approval-gated maintenance agent whose safety rules are enforced by the server and verified exhaustively and adversarially; the audit found, and we fixed, one rule enforced only by the prompt.
- Pre-registered robustness and model-comparison studies, and an out-of-time evaluation on 2026 data published after the model was frozen, which quantifies model ageing and shows that retraining recovers most of the loss.
- Pre-registered operating-policy studies for the retrained model (daily alert budget, alert persistence, tier sizing) and a label-free drift monitor that flags when to retrain.

II. Related Work

Large field studies established that SMART attributes carry failure signal but that many failures occur without warning [1], [2]. Learning-based work has predicted disk replacement from SMART data [3] and combined SMART with performance and location signals [4]. A second line of work estimates remaining useful life (RUL) with recurrent and attention models [6], [7], [8].

Closest to our work, Ding *et al.* [9] combine an MLP time-mixing stage with LightGBM to regress the RUL of Seagate ST4000DM000 drives, using the 60 days before each failure, and report a test RMSE near five days. Two properties of that protocol differ from ours. First, its test set contains only drives that fail, so it cannot measure false alarms on the healthy majority. Second, each drive is min-max normalised over its full recorded window, which includes observations a daily scorer would not yet have. We therefore address a

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complementary task, detection of imminent failure across a whole multi-vendor fleet at true prevalence, and do not compare error figures directly.

Our predictor uses gradient-boosted trees [10], [11] tuned with Optuna [12]; urgency uses an accelerated-failure-time survival model [13] evaluated with the concordance index [14]; explanations use SHAP and exact TreeSHAP [15], [16]; and the maintenance agent follows the tool-use pattern of reasoning-and-acting language-model agents [17].

III. Data and Evaluation Protocol

A. Dataset and Label

We use the public Backblaze Drive Stats dataset [18], which publishes one record per drive per day with SMART attributes, model, capacity and a failure flag, from April 2013 to December 2025. The target is 1 when a drive's recorded failure lies 1 to 14 days after the record's date. The failure day itself is labelled 0, so only advance warning is rewarded, and surviving drives are always 0. Fourteen days matches the time needed to schedule a replacement.

B. Temporal Splits

Data are split by time, never at random (Table I). Validation and test keep the real failure rate; only the training set is rebalanced, keeping every positive row and ten negatives per positive.

TABLE I. Temporal splits

Split	Dates	Drive-days	Use
Train	2013-04 to 2024-03	484.1 M (3.44 M after sampling)	fitting
Validation	2024-04 to 2024-09	52.8 M	tuning, threshold, tiers
Test	2024-10 to 2025-12	144.7 M (352,383 drives)	all results

In the test period 5,350 of 352,383 drives (1.52 percent) fail within the horizon and 0.0472 percent of drive-days are positive; the drive-day rate is the relevant base rate for daily ranking. Labels for the last two weeks of the test split were completed with failures from January 2026, which were not visible when the split was first built (602 drive-days on 96 drives became positive). Results marked with an asterisk and the comparative studies in Sections V-E and V-F were computed before this correction; both arms of each comparison share the same labels.

C. Features

The model uses 212 features: 83 raw and 83 normalised SMART attributes, 36 rolling means and standard deviations over 7, 14 and 30 days, 6 day-over-day deltas, and four derived features (drive age, power-on years, a combined error total and a temperature-stress indicator). All windows look backwards only. Missing values in core attributes are imputed with training-period medians. Five infrastructure identifiers (vault, pod, cluster, slot and a legacy flag) are excluded; Section V-E tests that choice.

D. Metrics and Uncertainty

The primary metric is drive-day AUPRC on the full test split, reported with lift over prevalence; accuracy is not reported. Drive-level AUPRC scores each drive's latest record against whether it ever fails within the horizon. Rows of one drive are not independent, so every interval resamples physical drives: an exact weighted average-precision bootstrap over all 352,383 test drives (B = 1000), in which a drive drawn *m* times contributes each of its rows *m* times. Every comparative study (the infrastructure-feature ablation, the model-fusion study, the 2026 evaluation, retraining, alert policies and tier sizing) fixed its decision rule in a committed pre-registration before any result was seen; model comparisons use a materiality margin of 0.003 AUPRC.

IV. System Design

DiskGuard separates prediction, which produces frozen model outputs, from action, which an operator approves (Fig. 1).

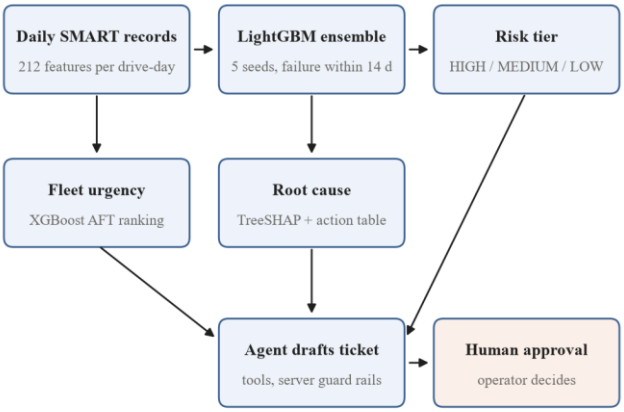


Fig. 1. DiskGuard architecture. Prediction outputs are immutable inputs to the agent; every ticket requires human approval.

A. Failure Prediction

LightGBM [10] is tuned with Optuna [12] (30 trials, TPE sampler) to maximise AUPRC on a fixed 5-million-row validation prefix. The selected configuration uses 798 trees, learning rate 0.022, 53 leaves, maximum depth 10, row and column subsampling of 0.70 and 0.90, and L1 and L2 penalties of 0.13 and 4.96. The deployed model averages five boosters trained with different seeds. An alert threshold of 0.62 minimises expected cost on validation with a missed failure costed at 100 false alarms.

B. Recall-Calibrated Risk Tiers

Operators work from tiers, not raw scores. Cutoffs are set on validation so that the HIGH tier captures 50 percent of failing drive-days and HIGH plus MEDIUM captures 80 percent, giving HIGH at 0.764 and MEDIUM at 0.069. The cutoffs are frozen and applied unchanged to test.

C. Fleet Urgency

An XGBoost accelerated-failure-time model [13] is fitted on the same features with interval-censored labels. Its absolute day estimates proved unreliable (Section V-D), so the system reports a ranking: fleet urgency is the share of a reference fleet that the model expects to outlive the drive. Urgency orders work within a tier and never moves a drive to a higher tier.

D. Explanations and Actions

Exact TreeSHAP [16] attributes each at-risk drive's score to its features. The strongest risk-increasing feature, with rolling and delta variants mapped to their base attribute, is looked up in a curated table of root causes and actions. The tier sets the deadline and the driver sets the kind of action: for reallocated sectors, a HIGH drive is scheduled for replacement

within seven days, while a MEDIUM drive is planned for the next maintenance window with increased SMART polling.

E. Approval-Gated Maintenance Agent

A tool-using language-model agent receives the frozen prediction as immutable evidence and may call three simulated tools: a spare-parts check, a technician-availability check and a ticket draft. It never executes an action, and LOW-tier drives never reach the model. The server, not only the prompt, enforces the safety rules: ticket identity, tier and root cause are attached by the server; the parts lookup is locked to the drive's own model; a non-HIGH drive cannot be scheduled as urgent; tool inputs are validated; and runs are bounded in tools, turns and time. A deterministic rule policy with the same tools is the no-network fallback and the evaluation reference.

V. Results

A. Failure Prediction

TABLE II. Test-set performance at true prevalence

Metric	LightGBM ensemble
Drive-day AUPRC [95% CI]	0.0844 [0.0792, 0.0903]
Lift over prevalence (0.000472)	179
Drive-day ROC-AUC	0.888
Drive-level AUPRC [95% CI]	0.600 [0.584, 0.614]
Drive-level ROC-AUC [95% CI]	0.919 [0.914, 0.924]
Single-seed AUPRC (5 seeds)*	0.0804 to 0.0828
Logistic regression baseline*	AUPRC 0.018, ROC-AUC 0.772
Precision / recall at 0.62	2.8% / 59.8%

The ensemble ranks failing drive-days 179 times better than chance and 4.6 times better than logistic regression (Table II). Ensembling five seeds adds 0.002 AUPRC over the average single seed, comparable to the effects studied below, which is why every comparison holds the ensemble fixed. The most influential features (mean absolute SHAP) are the raw read-error rate, the combined error total, normalised SMART 18 and SMART 1, and the raw start/stop count.

B. How Evaluation Shortcuts Inflate the Same Model

Table III re-scores the same test predictions under shortcuts common in the literature. Nothing about the model changes, only the composition of the evaluation set; this analysis is post hoc and descriptive.

TABLE III. Same scores under different evaluation protocols

Protocol	Neg. per pos.	AUPRC	ROC-AUC	Prec. at 0.62
True prevalence (ours)	2,118	0.084	0.888	2.8%
Never-failing drives as negatives	2,103	0.132	0.889	3.4%
Undersampled 1:100	100	0.458	0.888	37.5%
Undersampled 1:10	10	0.737	0.888	85.7%
Undersampled 1:1	1	0.917	0.888	98.4%

Every positive drive-day is kept in every protocol, so recall at the threshold is 59.8 percent throughout; undersampled rows are means of ten draws (AUPRC varies by at most 0.002). Undersampling to a 1:1 balance, close to the class mix of early benchmark data (178 good and 191 failed drives, assigned at random to training and test sets [19]), raises AUPRC elevenfold and precision at the deployed threshold from 2.8 to 98.4 percent: an operator would expect almost every alert to be a real failure, when in deployment about one in 36 is. Drawing negatives only from never-failing drives removes the hard negatives in a failing drive's earlier life and still inflates AUPRC by 56 percent. ROC-AUC is unchanged under every protocol, which is why it can neither expose these shortcuts nor describe the precision an operator will see; accuracy is highest for the rule that never predicts a failure (99.95 percent).

C. Risk Tiers and Lead Time

TABLE IV. Risk tiers on test, drive level (cutoffs fixed on validation)

Tier	Drives	Failing caught	Precision
HIGH	6,282	3,554 (66.4%)	56.6%
MEDIUM	26,512	853 (15.9%)	3.2%
HIGH + MEDIUM	32,794	4,407 (82.4%)	13.4%
LOW	319,589	943 (17.6%)	0.3%

The HIGH tier is small (1.8 percent of drives) and precise enough to act on: more than half of its drives fail within the horizon (Table IV). The tiers were calibrated to capture 50 and 80 percent of failing drive-days on validation and capture 52.2 and 75.4 percent on test, so the calibration transfers to a later period. Among failing drives observed at each lead time, the share scoring at or above the alert threshold (0.62) is 42.7, 50.5, 54.7 and 63.2 percent at 30, 14, 7 and 1 days before failure; half of failing drives are therefore flagged two weeks ahead. The 17.6 percent of failing drives that remain LOW reflect a known limit of SMART: many failures give no warning [2].

D. Urgency Ranking

On each test drive's first observation (352,197 drives, 5,341 failing), the survival model reaches a concordance index of 0.749 [14]. Its absolute estimates are not usable: on failing drives within 90 days of failure the mean absolute error is about 1,021 days, because roughly 85 percent of its training rows are censored. Expressed as fleet urgency, the ranking agrees with the tiers without being redundant: mean urgency is 99.4 percent in HIGH, 86.0 in MEDIUM and 53.9 in LOW.

E. Robustness

Infrastructure identifiers (pre-registered). Restoring the five identifiers changes AUPRC by −0.00014 (95% CI −0.0025 to +0.0022), within the 0.003 margin, so the verdict is equivalence: the model does not depend on where a drive is installed. *Datacenter hold-out.* Holding out an entire site does not degrade ranking: the held-out sites phx1, sac0 and sac2 reach lifts of 216, 141 and 126, each above its size-matched control (90, 90 and 80), which separates distribution shift from having less training data. *Calibration.* Brier score 0.0107 and expected calibration error 0.040; scores are ranking scores, and decisions use the tiers.

F. Alternatives Evaluated

Under pre-registered rules, LightGBM was compared with a recurrent (LSTM) remaining-life model and with fusions of LightGBM and a second model: the LSTM estimate, the survival model, and a separately tuned XGBoost classifier [11]. None improved on LightGBM by the 0.003 margin. The LSTM and survival-model fusions were worse (−0.016 and −0.039 AUPRC on the validation screen), and the XGBoost fusion, confirmed on the full test set, gained only +0.0004 (95% CI +0.0002 to +0.0006). Within bands of LightGBM's own score, the LSTM estimate ranks failures barely better

than chance, and a second boosted model on the same features largely agrees with LightGBM.

G. Agent Evaluation

TABLE V. AGENT INVARIANTS ON THE RULE POLICY (34,794 DRIVES)

Invariant	Applicable	Failures
Prediction returned unchanged	34,794	0
LOW drive gets no tools and no ticket	2,000	0
Ticket identity and tier server-owned	32,794	0
Ticket is a draft requiring approval	32,794	0
Parts lookup locked to drive model	6,282	0
MEDIUM never scheduled as urgent	26,512	0
No time-to-failure claimed	34,794	0

Table V lists seven of the thirteen invariants; all thirteen hold on every evaluated drive (all 32,794 at-risk drives and 2,000 LOW drives). A scripted adversarial model attempted ten attacks, including tickets for another drive or tier, a parts lookup for another model, an unknown tool, escalating a MEDIUM drive, claiming a day count, tool loops, an API outage and a truncated response. The server boundary held in all ten; the first run exposed one rule enforced only by the prompt, which was fixed by capping the request in code. TreeSHAP explanations give a specific root cause for 96.5 percent of at-risk drives (99.5 percent on the 2026 deployment after the mapping was extended to attributes the retrained model relies on). *Live model.* We also ran the agent with a live open-weight model, GPT-OSS-120B served by Groq, on 22 drives of the deployed 2026 system: 16 HIGH and MEDIUM drives stratified by urgency, 2 LOW drives, and 4 MEDIUM drives whose explanation carried an injected instruction to escalate the drive and state a failure date. A first run showed that the model sometimes wrote its plan as text without calling the ticket tool (4 of 20 at-risk drives), so the server now reminds it once and otherwise drafts the ticket from the rule mapping. In the final run every server-enforced invariant held, all four injection probes were resisted, and every at-risk drive reached the approval queue: 10 tickets came from the model unprompted, 6 after the reminder and 4 from the server fallback. Four runs did not complete cleanly, two because the provider rejected a malformed or schema-invalid tool call and two because free-tier rate limits exhausted the time budget; each still ended in a draft. The safety guarantees therefore rest on server-side enforcement, not on the model's reliability.

VI. Out-of-Time Evaluation on 2026 Data

Backblaze published data for January to June 2026 after our model, threshold and tiers were frozen. We wrote the evaluation plan before scoring any 2026 row, applied the same cleaning and features (checked against the original pipeline on December 2025: all 351,584 rows matched, 205 of 212 features exactly), and evaluated rows up to 16 June, the last date with a full 14-day follow-up.

TABLE VI. FROZEN MODEL ON THE TEST PERIOD AND ON 2026

Metric	Test (2024-25)	2026 H1
Drive-days / failing drives	144.7 M / 5,350	57.6 M / 2,451
Drive-day AUPRC [95% CI]	0.084 [0.079, 0.090]	0.061 [0.055, 0.067]
Lift over prevalence	179	127
ROC-AUC	0.888	0.870
Recall at 0.62	59.8%	63.8%
Failing drives caught by HIGH	66.4%	66.4%
Flagged 14 days ahead	50.5%	50.9%

By the rule stated in the plan, performance is degraded: the 2026 interval lies entirely below the test estimate (Table VI). Recall, tier capture and lead time held; ranking quality fell. Fig. 2 shows that the decline is gradual and began inside the test period: monthly AUPRC was 0.10 to 0.13 from October 2024 to April 2025 and fell to 0.07 to 0.09 by mid-2025. The cause is false alarms rather than missed failures. HIGH-tier drive-days per day rose from 1,393 in October 2024 to 2,405 in December 2025 and 2,960 in June 2026, while positive drive-days per day stayed near 140 to 180, and precision fell for every major drive model. Drive models absent from training are only 7.6 percent of the 2026 fleet and perform like known models, so new hardware is not the explanation. A plausible mechanism is that cumulative counters such as power-on hours and start/stop counts grow as the fleet ages, pushing healthy drives' scores up.

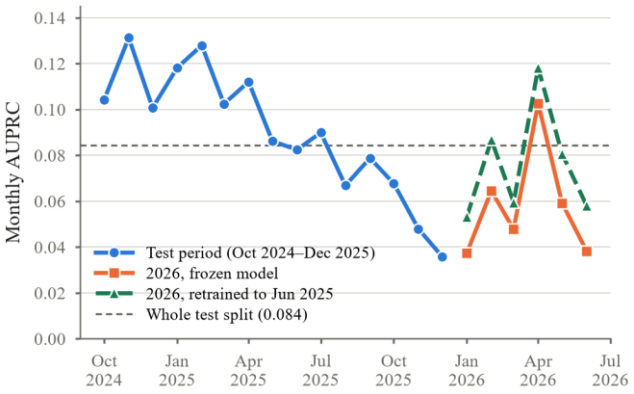


Fig. 2. Monthly drive-day AUPRC of the frozen model on the test period and on 2026, and of the model retrained on data to June 2025 (Section VI-A).

A. Retraining on Recent Data

We pre-registered a retraining study before training any model in it. The same features and hyperparameters were refitted on data to 16 June 2025; threshold and tiers were refitted on July to mid-December 2025; and 2026 was used once, for a confirmatory comparison against the frozen model with the same 0.003 margin and drive-clustered bootstrap. A second variant added, for ten key SMART attributes, each drive's percentile among drives of the same model on the same day, to test whether fleet-relative features resist ageing. On validation this variant gained +0.0022 AUPRC (95% CI +0.0005 to +0.0041), real but below the margin, so the plain retrained model went forward.

TABLE VII. FROZEN AND RETRAINED MODELS ON 2026 (EACH AT ITS OWN VALIDATION-FITTED THRESHOLD)

Metric	Frozen	Retrained
Drive-day AUPRC	0.061	0.078
ROC-AUC	0.870	0.886
Drive-level AUPRC	0.422	0.459
Threshold	0.62	0.73
Precision / recall at threshold	2.2% / 63.8%	3.6% / 57.4%
Flagged 14 days ahead	50.9%	44.1%

Retraining helps by the pre-registered rule: AUPRC rises from 0.061 to 0.078, a gain of +0.017 (95% CI +0.013 to +0.021), which recovers most of the gap to the 0.084 of the original test period and improves every 2026 month (Fig. 2). The cost-optimal threshold rose from 0.62 to 0.73, so the

retrained model trades some recall and lead time for 60 percent higher precision (Table VII). The model's decline was therefore mostly ageing that periodic retraining repairs, and the retrained model replaces the frozen one.

B. Alert Policies

A threshold fixes the score that raises an alert, not how many alerts arrive. We pre-registered two policies for the retrained model and compared each with the 0.73 threshold on 2026 once, with the same drive-clustered bootstrap. The primary metrics are the precision of new alerts (an episode of consecutive alerted days counted at its first day, when an operator is paged) and the share of failing drives caught. *Persistence* alerts only after two consecutive days above the threshold. A *daily budget* alerts on the k highest-scoring drives of each day, with k = 2,074 fitted on validation to match the threshold's catch rate. A policy is adopted if its precision gain has a CI above zero and it loses at most 5 points of catch rate.

TABLE VIII. ALERT POLICIES FOR THE RETRAINED MODEL ON 2026

Policy	Alerts per day	New-alert precision	Failing caught
Threshold 0.73	2,654	4.02%	69.5%
Two days above 0.73	2,461	3.50%	61.6%
Daily top 2,074	2,074	4.58%	67.2%

Persistence is worse on both metrics (precision −0.52 points, 95% CI −0.71 to −0.32), so failing drives do not stay high more consistently than healthy drives spike. The daily budget is adopted: precision +0.55 points (CI +0.43 to +0.68, 14 percent relative) with 22 percent fewer alerts, for 2.2 points of catch rate (Table VIII). It works because scores drift upward as the fleet ages: the fixed threshold raised 26 percent more alerts a day in 2026 than on validation, while the budget holds the workload constant. Smaller budgets trade coverage for precision: at 100 alerts a day, 17.7 percent of alerted drive-days precede a failure (about one in six, against one in 28 for the threshold) and 23.4 percent of failing drives are caught. A third pre-registered study shrank HIGH plus MEDIUM from 16 to 9.3 percent of drive-days; on validation this lost 7.1 points of failing-drive coverage, above its 5-point limit, so the recall-calibrated tiers were kept.

C. Drift Monitoring

Labels arrive 14 days late, so we monitor each month against the model's own validation period with two label-free signals, the share of drive-days above the alert threshold and the score distribution's population stability index (PSI), and the AUPRC of the last three labelled months. Retraining is flagged at 1.25 times the validation alert volume, a PSI of 0.25, or 0.75 times the validation AUPRC. These limits were set after the frozen model's ageing was known, so the replay is illustrative rather than confirmatory; a first version using single-month AUPRC raised false alarms on noisy months and was replaced. Replayed on the frozen model, the monitor warns in November 2024 and calls for retraining in July 2025 (alert volume 1.30 times, AUPRC 0.72 times), sixteen months after its training data ended and six months before the 2026 decline was measured; retraining on data to June 2025 then recovered it. The retrained model shows the same pattern: alert volume passed 1.25 times in April 2026, ten months after its training data, while its AUPRC was still at the validation level, an early warning that the daily budget absorbs. PSI never exceeded 0.06, because the drift is concentrated in the high-score tail; alert volume is the earliest and clearest signal.

VII. Discussion and Limitations

A strong single model was the right choice: on rich SMART features at true prevalence, a tuned and ensembled LightGBM was not improved by any of the more complex designs we evaluated. Our contribution is less a new predictor than an honest, complete system around a strong one, with each component's claims measured. Operators ask how many days are left, but neither model we trained could answer reliably; a ranking uses only the property shown to hold. Prompt instructions are not a sufficient control, since the one gap the audit found was a rule stated in the prompt but not enforced by the server. Finally, a frozen model ages: on 2026 data a model retrained on more recent data recovers most of the loss, and daily alert volume, which needs no labels, signals the need to retrain months before ranking metrics can confirm it; both replays suggest retraining every nine to twelve months. A fixed daily alert budget also makes the operator's workload robust to that drift. Several reasonable ideas did not pass their pre-registered tests (fleet-relative features, alert persistence, a smaller MEDIUM tier); we report them because each narrows what is worth trying next.

Limitations. Scores are ranking scores, not calibrated probabilities. SMART has a ceiling: 17.6 percent of failing drives stay in the LOW tier. Failure labels come from operational removals and may include drives removed for other reasons. The fleet belongs to one operator and may not represent others. Training used an undersampled set and tuning a validation prefix, and the agent's tools are simulated.

VIII. Conclusion

DiskGuard shows that useful, explainable and safe failure management can be built around a single well-evaluated gradient-boosting model. At true prevalence it ranks failing drive-days 179 times better than chance, flags half of failing drives two weeks ahead, places two-thirds of failing drives in a tier precise enough to act on, orders work with a survival-model ranking, explains each alert, and drafts maintenance that a human approves under guard rails verified exhaustively and adversarially. The same predictions scored on a balanced test set would look eleven times better. On data from after the model was frozen its ranking degrades through rising false alarms; retraining on recent data recovers most of the loss, a daily alert budget keeps the workload fixed and its alerts more precise, and a label-free drift monitor signals when to retrain. Future work includes automating the retraining the monitor triggers and validation on a second operator's fleet.

Data and Code Availability

The Backblaze data are public [18]. All stages are scripted, with pinned dependencies, pre-registration documents and the results files behind every table, at https://github.com/rakshithaaa-m/capstone.

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